Project Planning & Management Document

1. Project Overview

- **INFERNOTRON**
- **Ojaswi Bajpai – Tech, Mechanisms and Coding,Testing,Software,Documentation**
  Aradhya Mohan – Mechanisms, Aesthetics, Testing,Documentation

- *Infernotron is a Bluetooth-controlled robotic car powered by an ESP32 microcontroller that can be operated through a mobile app. It can move in different directions, detect objects using an IR sensor, and activate laser targeting when an object is detected. The system also simulates a shooting effect using LEDs and a buzzer, creating an interactive robotics and security-prototype experience.*

2. Team Roles & Responsibilities

Member Name	Role	Responsibilities
Ojaswi Bajpai	Software Developer and Systems Integrator responsible for programming the ESP32, integrating electronic components, and coordinating the overall technical implementation of the robot.	⍰ Developed and implemented the ESP32 control code ⍰ Handled hardware setup and circuit connections ⍰ Procured electronic components and materials ⍰ Assisted in designing mechanical mechanisms ⍰ Prepared project documentation ⍰ Conducted system testing and troubleshooting

		⍰ Managed software integration between the robot and the mobile application
Aradhya Mohan	Design and Mechanisms Lead responsible for the physical structure, visual design, and mechanical assembly of the robot.	⍰ Designed the external appearance and aesthetic elements of the robot ⍰ Assisted in building and assembling the mechanical structure ⍰ Helped in sourcing and organizing required materials ⍰ Participated in system testing to evaluate performance and stability

Clearly define who is responsible for what (software, electronics, documentation, etc.)

3. Project Timeline & Milestones

Week	Goals	Deliverables	Status
Week 0	• Brainstorm and finalize the concept for the robotics project • Define the purpose and core functionality of the robot • Get project approval from the instructor • Plan the required components, software, and system structure	⍰ Finalized project idea (**Infernotron – Bluetooth controlled robotic car**) ⍰ Initial system concept including movement, object detection, and shooting effect ⍰ List of required components (ESP32, motors, motor driver, IR sensor, lasers, LEDs, buzzer, battery) ⍰ Basic project plan and approval	⍰ Project concept successfully finalized and approved ⍰ Team roles discussed and assigned ⍰ Required components identified and prepared for procurement

Week 1	⍰ Build the initial prototype of the robot ⍰ Write the first version of the ESP32 control code ⍰ Establish Bluetooth communication with the mobile application ⍰ Test basic motor movement and component functionality	⍰ First working prototype of the robot hardware ⍰ Initial MicroPython code for ESP32 ⍰ Mobile app interface developed using **MIT App Inventor** ⍰ Basic movement control and communication between the app and the robot	⍰ Prototype assembled and components connected ⍰ Bluetooth communication successfully established ⍰ Basic movement commands working ⍰ Initial testing completed with minor debugging

4. Weekly Progress Log

Week 0

Planned Tasks

- Brainstorm project ideas
- Finalize the concept and functionality
- Define system components and architecture
- Obtain instructor approval

Completed Tasks

- Selected the project idea of building **Infernotron**, a Bluetooth-controlled robotic car
- Discussed system features such as object detection, laser targeting, and shooting effects
- Finalized the list of hardware components required
- Received approval to proceed with the project

Challenges Faced

- Choosing a concept that was both feasible and interesting
- Determining the components needed for the robot system

Solutions

- Researched existing robotics projects for inspiration
- Evaluated available hardware and selected components that could achieve the intended functionality

Week 1

Planned Tasks

- Procure the required components
- Assemble the robot chassis and electronic components
- Write initial ESP32 code
- Establish Bluetooth communication with the mobile app

Completed Tasks

- Assembled the hardware including ESP32, motor driver, motors, and sensors
- Developed the first version of the ESP32 control code
- Built the mobile app interface for robot control
- Successfully connected the robot to the app via Bluetooth and tested basic movement

Challenges Faced

- Initial Bluetooth connection issues between the app and ESP32
- Wiring complexity while integrating multiple components

Solutions

- Debugged the code and adjusted Bluetooth settings to establish a stable connection
- Reorganized the wiring and verified connections to ensure proper hardware functionality.